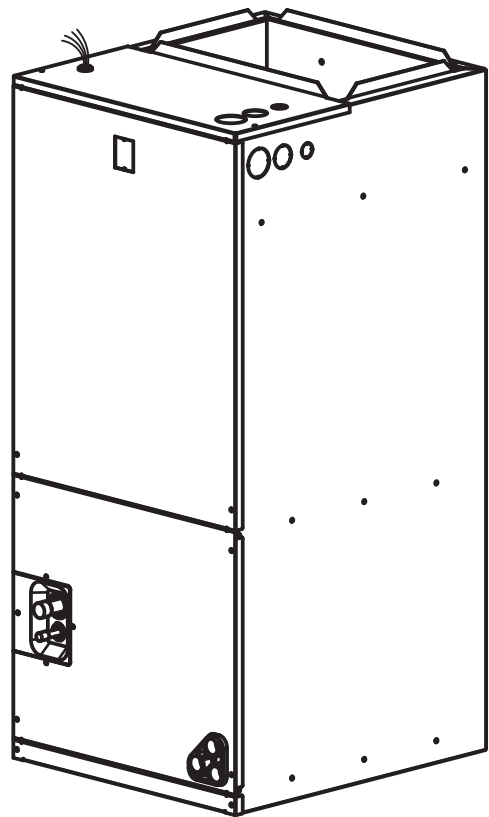


# INSTALLATION INSTRUCTIONS

## HIGH EFFICIENCY AIR HANDLERS

1.5-5Tons

FEATURING R-410A OR R22 REFRIGERANT



RECOGNIZE THIS SYMBOL AS AN INDICATION OF IMPORTANT SAFETY INFORMATION



### WARNING

These instructions are intended as an aid to qualified licensed service personnel for proper installation, adjustment and operation of this unit. Read these instructions thoroughly before attempting installation or operation. Failure to follow these instruction may result in improper installation, adjustment, service or maintenance possibly resulting in fire, electrical shock, property damage, personal injury or death.



DO NOT DESTROY THIS MANUAL

Please read carefully and keep in a safe place for future reference by a serviceman.

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This document is customer property and is to remain with this unit.  
These instructions do not cover all the different variations systems nor does it provide for every possible contingency to be met in connection with installation.  
All phases of this installation must comply with NATIONAL STATE AND LOCAL CODES. If additional information is required please contact your local distributor.

## 1.0 SAFETY



This is a safety alert symbol. When you see this symbol on labels or in manuals, be alert to the potential for personal injury.



This is an attention alert symbol. When you see this symbol on labels or in manuals, be alert to the potential for personal injury.



### WARNING

Disconnect all power to unit before installing or servicing. More than one disconnect switch may be required to de-energize the equipment. Hazardous voltage can cause severe personal injury or death.



### WARNING

If removal of the blower assembly is required, all disconnect switches supplying power to the equipment must be de-energized and locked (if not in sight of unit ) so the field power wires can be safely removed from the blower assembly. Failure to do so can cause electrical shock resulting in personal injury or death.



### WARNING

Because of possible damage to equipment or personal injury, installation, service, and maintenance should be performed by a trained, qualified service personnel. Consumer service is recommended only for filter cleaning / replacement. Never operate the unit with the access panels removed.



### WARNING

These instructions are intended as an aid to qualified, licensed service personnel for proper installation, adjustment and operation of this unit. Read these instructions thoroughly before attempting installation or operation. Failure to follow these instructions may result in improper installation, adjustment, service or maintenance possibly resulting in fire, electrical shock, property damage, personal injury or death.



## WARNING

The unit must be permanently grounded. Failure to do so can result in electrical shock causing personal injury or death.



## WARNING

**PROPOSITION 65:** This appliance contains fiberglass insulation. Respirable particles of fiberglass are known to State of California to cause cancer.

All manufacturer products meet current federal OSHA Guidelines for safety. California Proposition 65 warnings are required for certain products, which are not covered by the OSHA standards.

California's Proposition 65 requires warnings for products sold in California that contain or produce any of over 600 listed chemicals known to the State of California to cause cancer or birth defects such as fiberglass insulation, lead in brass, and combustion products from natural gas.

All "new equipment" shipped for sale in California will have labels stating that the product contains and /or produces Proposition 65 chemicals. Although we have not changed our processes, having the same label on all our products facilitates manufacturing and shipping. We cannot always know "when, or if" products will be sold in the California market.

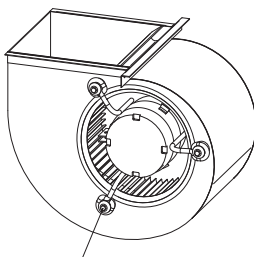
You may receive inquiries from customers about chemicals found in, or produced by, some of our heating and air-conditioning equipment, or found in natural gas used with some of our products. Listed below are those chemicals and substances commonly associated with similar equipment in our industry and other manufacturers.

Glass Wool (Fiberglass) Insulation  
Carbon Monoxide (CO).  
Formaldehyde  
Benzene

More details are available at the websites for OSHA (Occupational Safety and Health Administration), at [www.osha.gov](http://www.osha.gov) and the State of California's OEHHA (Office of Environmental Health Hazard Assessment), at [www.oehha.org](http://www.oehha.org). Consumer education is important since the chemicals and substances on the list are found in our daily lives. Most consumers are aware that products present safety and health risks, when improperly used, handled and maintained.



## CAUTION



BLOWER MOTOR SHIPPING BOLT

Make sure the blower motor support is tight (3-motor mount bolts) then check to see if wheel is secured to motor shaft before operating unit.



## WARNING

The first 6 inches of supply air plenum and ductwork must be constructed of sheet metal as required by NFPA 90B. The supply air plenum or duct must have a solid sheet metal bottom directly under the unit with no openings, registers or flexible air ducts located in it. If flexible supply air ducts are used they may be located only in the vertical walls of rectangular plenum, a minimum of 6 inches from the solid bottom. Metal plenum or duct may be connected to the combustible floor base, if not, it must be connected to the unit supply duct exposed to the supply air opening from the downflow unit. Exposing combustible (non-metal) material to the supply opening of a downflow unit can cause a fire resulting in property damage, personal injury or death.

Exception warning to downflow:

Installations on concrete floor slab with supply air plenum and ductwork completely encased must be not less than 2 inches of concrete (See NFPA 90A).

## 2.0 GENERAL

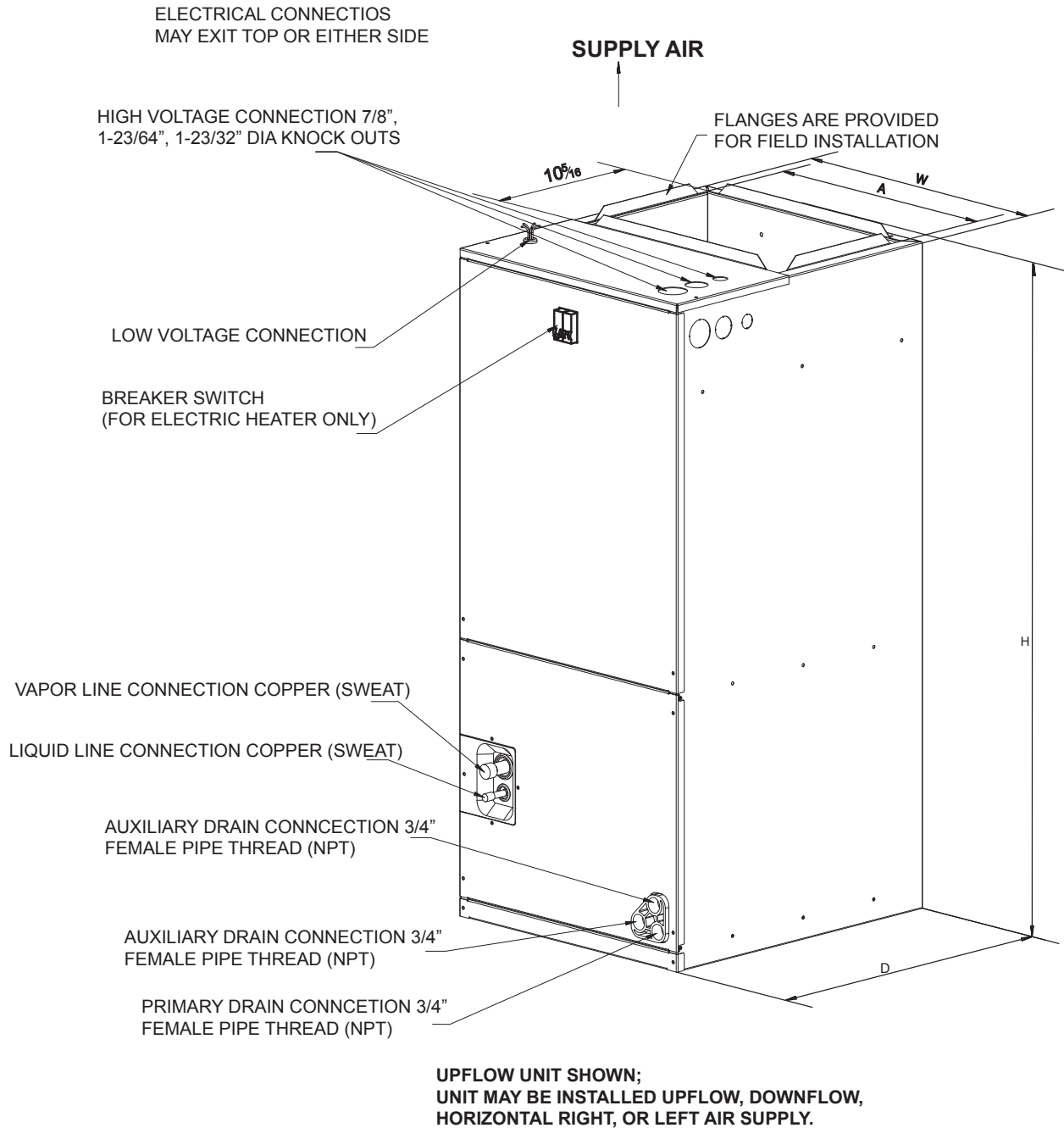
These Air Handlers are four way compatible for upflow ,downflow ,horizontal right and left positions.

This Air Handler provides the flexibility for installation in any upflow or downflow horizontal application. The direct drive motors provides a selection of air volume to match any application. 3-Speed motors provide selections of air flow to meet desired applications.

Top and side power and control wiring, accessible screw terminals for control wiring all combine to make the installation easy, and minimize installation cost. Please contact your local distributor. See fig.1.

## 2.1 UNIT DIMENSIONS

NOTE: 24" CLEARANCE IS REQUIRED IN THE FRONT OF THE UNIT FOR FILTER AND COIL MAINTENANCE.



### DIMENSIONAL DATA

Fig.1 DIMENSIONS

| MODEL SIZE | Dimensions inch [mm]        |                           |                            |                    | UNIT WEIGHT<br>/SHIPPING WEIGHT<br>(LBS./[kg]) |
|------------|-----------------------------|---------------------------|----------------------------|--------------------|------------------------------------------------|
|            | UNIT HEIGHT<br>"H" IN. [mm] | UNIT WIDTH<br>"W" IN.[mm] | UNIT LENGHT<br>"D" IN.[mm] | SUPPLY<br>DUCT "A" |                                                |
| 18         | 41-3/8"[1050]               | 18-1/8"[ 460]             | 20-1/2"[520]               | 16"[406]           | 103/116 [47]/[53]                              |
| 24         | 41-3/8"[1050]               | 18-1/8"[ 460]             | 20-1/2"[520]               | 16"[406]           | 106/119 [48]/[54]                              |
| 30         | 41-3/8"[1050]               | 18-1/8"[ 460]             | 20-1/2"[520]               | 16"[406]           | 110/123 [50]/[56]                              |
| 36         | 46-1/2"[1180]               | 19-5/8"[500]              | 21-5/8"[550]               | 18"[456]           | 134/147 [61]/[67]                              |
| 42         | 46-1/2"[1180]               | 19-5/8"[500]              | 21-5/8"[550]               | 18"[456]           | 134/147 [61]/[67]                              |
| 48         | 54-1/2"[1385]               | 22"[560]                  | 24"[610]                   | 19-1/2"[496]       | 169/185 [77]/[84]                              |
| 60         | 54-1/2"[1385]               | 22"[560]                  | 24"[610]                   | 19-1/2"[496]       | 172/187 [78]/[85]                              |

## 3.0 APPLICATIONS

### 3.1 VERTICAL UPFLOW

- Vertical Upflow configuration is the factory set on all models (see Fig 1).
- If return air is to be ducted, install duct flush with floor. Use fireproof resilient gasket 1/8 to 1/4 in. thick between the ducts, unit and floor. Set unit on floor over opening.

#### IMPORTANT NOTE

Torque applied to drain connections should not exceed 15.ft.lbs.(see Fig.1&2)

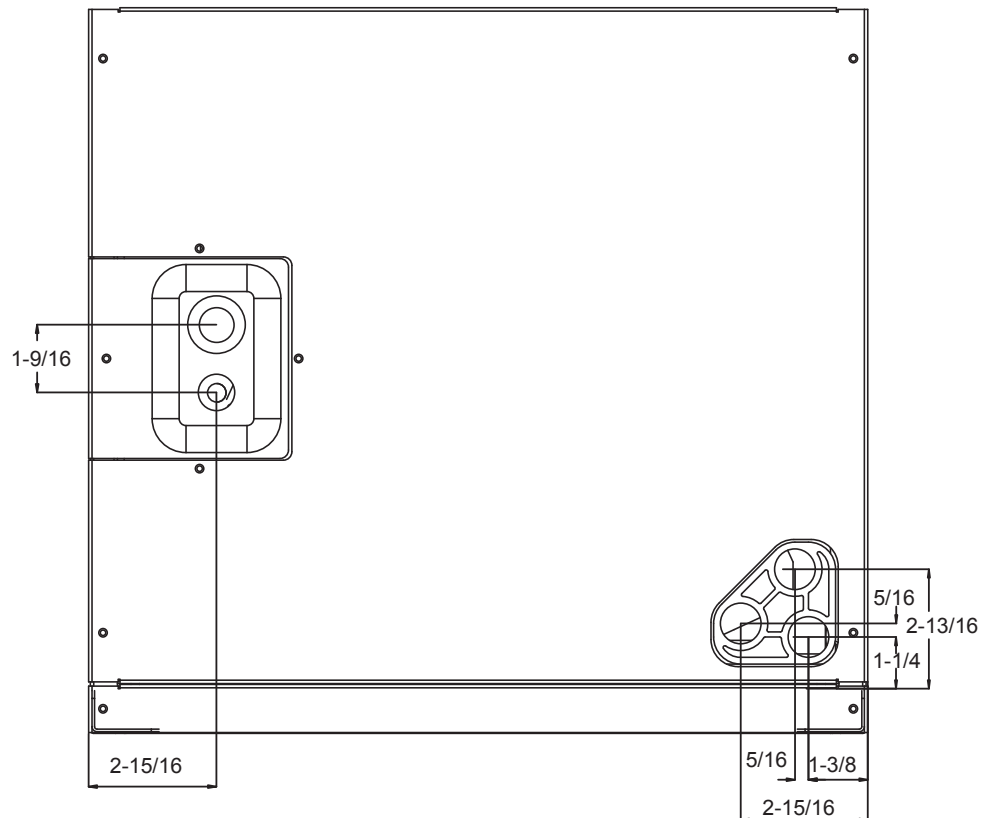


Fig.2 DIMENSIONS FOR FRONT CONNECT COIL

### 3.2 VERTICAL DOWNFLOW

**Conversion to Vertical Downflow:** A vertical upflow unit may be converted to vertical downflow . Remove the door and indoor coil and reinstall 180° from original position. See Fig3.

**IMPORTANT:** To comply with certification agencies and the National Electric Code for horizontal right application, the circuit breaker(s) on field-installed electric heater kits must be re-installed per procedure below so that the breaker switch “on” position and marking is up and, “off” position and marking is down.

- To rotate breaker(s): Rotate one breaker set (circuit) at a time starting with the one on the right. Loosen both lugs on the load side of the breaker. (Make sure that wires are identified and are reinstalled into proper breaker). Wires are bundles with wire ties, one bundle going to the right lug and one bundle going to the left lug.

- Using a screwdriver or pencil, lift blue plastic tab with hole away from breaker until breaker releases from mounting opening.
- With breaker held in hand, rotate breaker so that “on” position is up, “off” position is down with unit in planned vertical mounting position. Insert right wire bundle into top right breaker lug, ensuring all strands of all wires are inserted fully into lug, and no wire insulation is in lug.
- Tighten lug as tight as possible while holding circuit breaker. Check wires and make sure each wire is secure and none are loose. Repeat for left wire bundle in left top circuit breaker lug.
- Replace breaker by inserting breaker mounting tab opposite white pull tab in opening, hook mounting tab over edge in opening.
- With screwdriver or pencil, pull blue tab with hole away from breaker while setting that side of breaker into opening. When breaker is in place, release tab, locking circuit breaker into location in opening.
- Repeat above operation for remaining breaker(s) (if more than one is provided).
- Replace single point wiring jumper bar, if it is used, on line side of breaker and tighten securely.
- Double check wires and lugs to make sure all are secure and tight. Check to make sure unit wiring to circuit breaker load lugs match that shown on the unit wiring diagram.



#### CAUTION

When using the unit with electrical heater, the switch is used only for electrical heater on the front of panel.

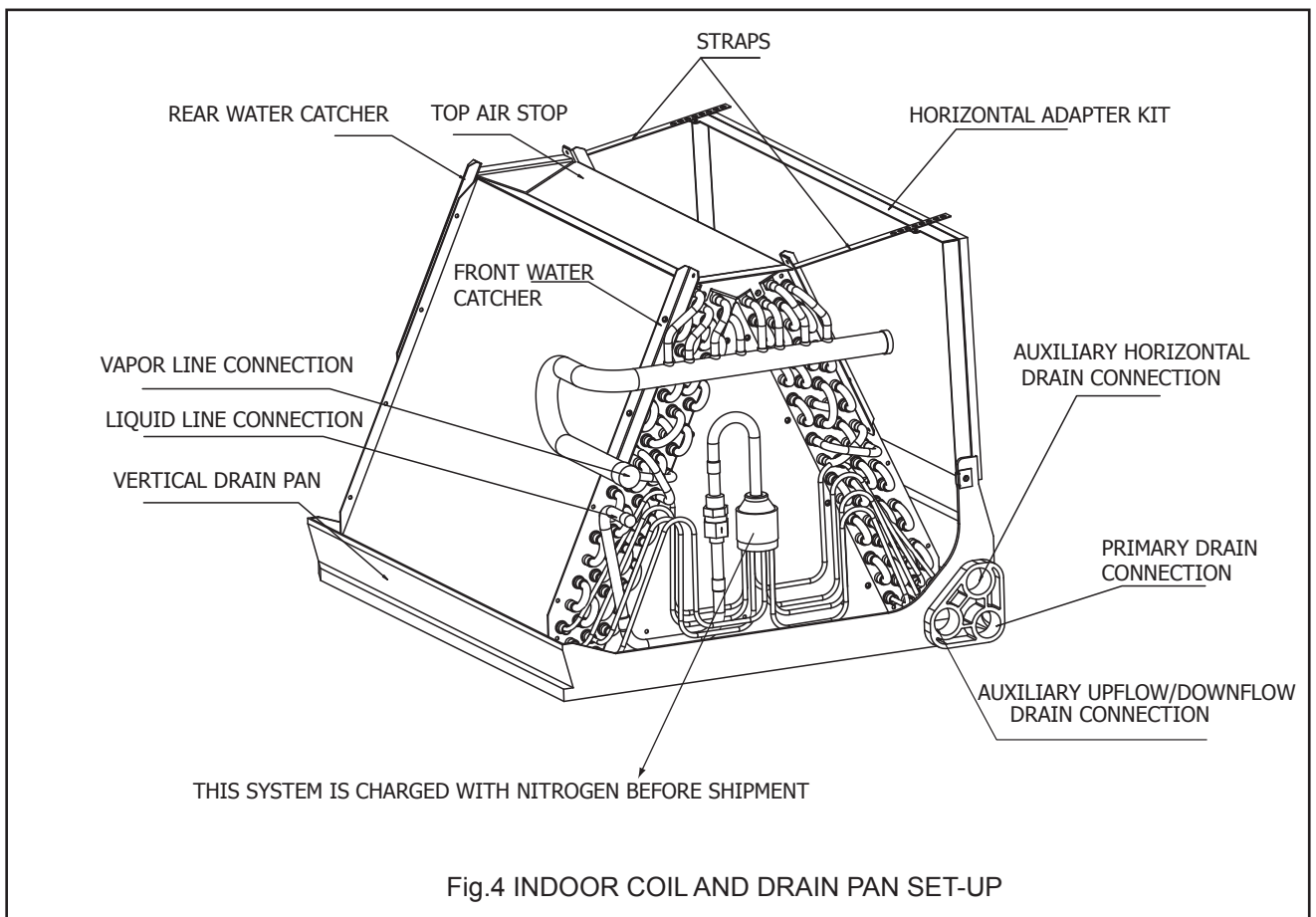
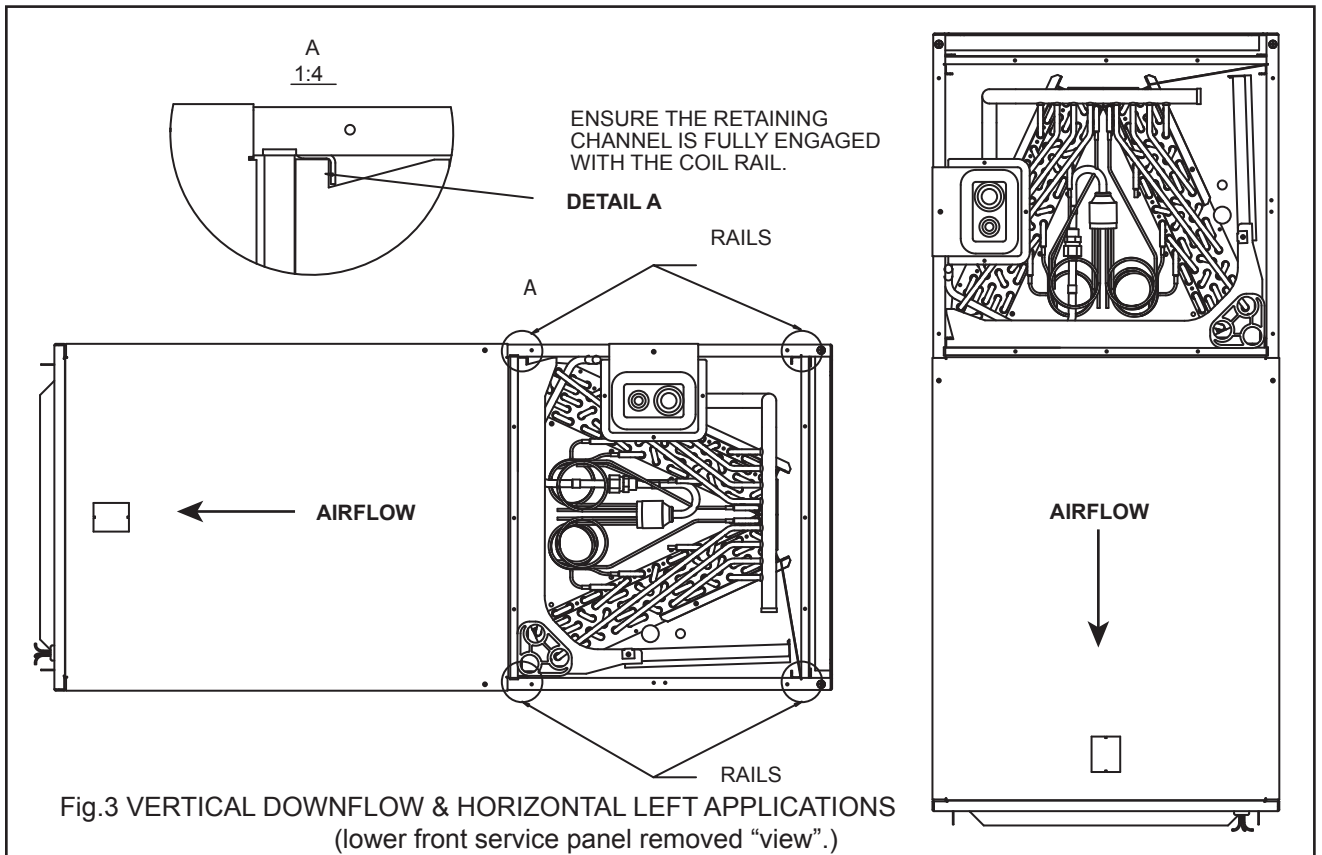
### 3.3 HORIZONTAL

Horizontal right is the default factory configuration for the units.

Horizontal left isn't the default factory configuration for the units.

Conversion to Horizontal: A vertical upflow unit may be converted to horizontal left by removing indoor coil assembly and reinstalling coil as shown for left hand air supply. And reinstall coil in unit as shown for left hand air supply.

- Rotate unit into the downflow position, with the coil compartment on top and the blower compartment on bottom. See Fig. 3.
- Reinstall the indoor coil 180° from original position. Ensure the retaining channel is fully engaged with the coil rail. See Fig. 3.
- Secondary drain pan kits are recommended when the unit is configured for the horizontal position over a finished ceiling and/or living space.





## CAUTION

Horizontal units must be configured for right hand air supply or left hand air supply. Horizontal drain pan must be located under indoor coil. Failure to use the drain pan can result in property damage.

**Conversion in Horizontal Direction:** Horizontal right-hand supply can be changed to horizontal left-hand supply by removing the indoor coil and reinstalling 180° from original.

### 3.4 INSTALLATION IN AN UNCONDITIONED SPACE

**IMPORTANT:** There are two pairs of coil rails in the air handler for default and counter flow application. If the air handler is installed in an unconditioned space, the two unused coil rails should be removed to minimize air handler surface sweating. The coil rails can be easily removed by taking off the 6 mounting screws from both sides of the cabinet.

## 4.0 ELECTRICAL WIRING

Field wiring must comply with the National Electric Code (C.E.C. in Canada) and any applicable local ordinance.



## WARNING

Disconnect all power to unit before installing or servicing. More than one disconnect switch may be required to de-energize the equipment. Hazardous voltage can cause severe personal injury or death.

### 4.1 POWER WIRING

It is important that proper electrical power is available for connection to the unit model being installed. See the unit nameplate, wiring diagram and electrical data in the installation instructions.

- If required, install a branch circuit disconnect of adequate size, located within sight of, and readily accessible to the unit.
- **IMPORTANT:** After the Electric Heater is installed, units may be equipped with one, two, or three 30/60 amp. circuit breakers. These breaker(s) protect the internal wiring in the event of a short circuit and serve as a disconnect. Circuit breakers installed within the unit do not provide over-current protection of the supply wiring and therefore may be sized larger than the branch circuit protection.
- Supply circuit power wiring must be 75°C minimum copper conductors only. See Electrical Data In this section for ampacity, wire size and circuit protector requirement. Supply circuit protective devices may be either fuses or "HACR" type circuit breakers.
- Power wiring may be connected to either the right, left side or top. Three 7/8", 1-3/8", 1-3/4" dia. concentric knockouts are provided for connection of power wiring to unit.
- Power wiring is connected to the power terminal block in unit electric cabinet.

## 4.2 CONTROL WIRING

**IMPORTANT:** Class 2 low voltage control wiring should not be run in conduit with main power wiring and must be separated from power wiring, unless class 1 wire of proper voltage rating is used.

- Low voltage control wiring should be 18 Awg. color-coded. For lengths longer than 100 ft., 16 Awg. wire should be used.
- Low voltage control connections are made to low voltage pigtails extending from top of air handler (upflow position - see Figure 3). Connections for control wiring are made with wire nuts. Control wiring knockouts (5/8 and 7/8) are also provided on the right and left side of the unit for side connection.
- See wiring diagrams attached to indoor and outdoor sections to be connected.
- Make sure, after installation, separation of control wiring and power wiring has been maintained.

## 4.3 GROUNDING



### WARNING

The unit must be permanently grounded. Failure to do so can result in electrical shock causing personal injury or death.

- Grounding may be accomplished by grounding metal conduit when installed in accordance with electrical codes to the unit cabinet.
- Grounding may also be accomplished by attaching ground wire(s) to ground lug(s) provided in the unit wiring compartment.
- Ground lug(s) are located close to wire entrance on left side of unit (up-flow). Lug(s) may be moved to marked locations near wire entrance on right side of unit (upflow). If alternate location is more convenient.
- Use of multiple supply circuits require grounding of each circuit to lug(s) provided in unit.

## 4.4 ELECTRICAL DATA

| MODEL | VOLTAGE | HERTZ | HP  | RPM | SPEEDS | CIRCUIT AMPS. | MAXIMUM CIRCUIT PROTECTOR |
|-------|---------|-------|-----|-----|--------|---------------|---------------------------|
| 18    | 208/230 | 60    | 1/8 | 580 | 3      | 0.68          | 15(A)                     |
| 24    | 208/230 | 60    | 1/5 | 701 | 3      | 0.95          | 15(A)                     |
| 30    | 208/230 | 60    | 1/4 | 872 | 3      | 1.48          | 15(A)                     |
| 36    | 208/230 | 60    | 1/3 | 794 | 3      | 1.63          | 15(A)                     |
| 42    | 208/230 | 60    | 1/2 | 882 | 3      | 1.8           | 15(A)                     |
| 48    | 208/230 | 60    | 1/2 | 845 | 3      | 2.11          | 15(A)                     |
| 60    | 208/230 | 60    | 3/4 | 963 | 3      | 2.79          | 15(A)                     |

## 4.5 ELECTRIC KIT MCA/MOP DATA

| Heat Kit Model | Air Handler Model | (kW)Electric Heat | MIN. Circuit Ampacity |            | MAX.Fuse or Breaker (HACR) Ampacity |       | Fan speed (AC/HP) |        |      |
|----------------|-------------------|-------------------|-----------------------|------------|-------------------------------------|-------|-------------------|--------|------|
|                |                   |                   | 240                   | 208        | 240                                 | 208   | Low               | Medium | High |
| MAYHTR1A05BKR* | 18                | 5                 | 27                    | 23.5       | 30                                  | 25    | ●                 | ●      | ●    |
| MAYHTR1A08BKR* |                   | 7.5               | 40                    | 34.8       | 45                                  | 35    | --                | ●      | ●    |
| MAYHTR1A10BKR* |                   | 10                | 53                    | 46.1       | 60                                  | 50    | --                | --     | ●    |
| MAYHTR1A05BKR* | 24                | 5                 | 27.3                  | 23.9       | 30                                  | 25    | ●                 | ●      | ●    |
| MAYHTR1A08BKR* |                   | 7.5               | 40.4                  | 35.2       | 45                                  | 40    | ●                 | ●      | ●    |
| MAYHTR1A10BKR* |                   | 10                | 53.4                  | 46.4       | 60                                  | 50    | --                | ●      | ●    |
| MAYHTR1A05BKR* | 30                | 5                 | 28                    | 24.5       | 30                                  | 25    | ●                 | ●      | ●    |
| MAYHTR1A08BKR* |                   | 7.5               | 41                    | 35.8       | 45                                  | 40    | ●                 | ●      | ●    |
| MAYHTR1A10BKR* |                   | 10                | 54                    | 47.1       | 60                                  | 50    | ●                 | ●      | ●    |
| MAYHTR1A05BKR* | 36                | 5                 | 28.3                  | 24.9       | 30                                  | 25    | ●                 | ●      | ●    |
| MAYHTR1A08BKR* |                   | 7.5               | 41.4                  | 36.2       | 45                                  | 40    | ●                 | ●      | ●    |
| MAYHTR1A10BKR* |                   | 10                | 54.4                  | 47.4       | 60                                  | 50    | ●                 | ●      | ●    |
| MAYHTR1A15BKR* |                   | 15                | 54.4/26               | 47.34/22.8 | 60/30                               | 50/25 | --                | ●      | ●    |
| MAYHTR1A20BKR* |                   | 20                | 54.4/52.1             | 47.4/45.1  | 60/60                               | 50/50 | --                | --     | ●    |
| MAYHTR1A05BKR* | 42                | 5                 | 28.3                  | 24.9       | 30                                  | 25    | ●                 | ●      | ●    |
| MAYHTR1A08BKR* |                   | 7.5               | 41.4                  | 36.2       | 45                                  | 40    | ●                 | ●      | ●    |
| MAYHTR1A10BKR* |                   | 10                | 54.4                  | 47.4       | 60                                  | 50    | ●                 | ●      | ●    |
| MAYHTR1A15BKR* |                   | 15                | 54.4/26               | 47.4/22.8  | 60/30                               | 50/25 | --                | ●      | ●    |
| MAYHTR1A20BKR* |                   | 20                | 54.4/52.1             | 47.4/45.1  | 60/60                               | 50/50 | --                | --     | ●    |
| MAYHTR1A05BKR* | 48                | 5                 | 28.8                  | 25.4       | 30                                  | 30    | ●                 | ●      | ●    |
| MAYHTR1A08BKR* |                   | 7.5               | 41.9                  | 36.7       | 45                                  | 40    | ●                 | ●      | ●    |
| MAYHTR1A10BKR* |                   | 10                | 54.9                  | 47.9       | 60                                  | 50    | ●                 | ●      | ●    |
| MAYHTR1A15BKR* |                   | 15                | 54.9/26               | 47.9/22.8  | 60/30                               | 50/25 | ●                 | ●      | ●    |
| MAYHTR1A20BKR* |                   | 20                | 54.9/52.1             | 47.9/45.1  | 60/60                               | 50/50 | ●                 | ●      | ●    |
| MAYHTR1A05BKR* | 60                | 5                 | 29.6                  | 26.1       | 30                                  | 30    | ●                 | ●      | ●    |
| MAYHTR1A08BKR* |                   | 7.5               | 42.6                  | 37.4       | 45                                  | 40    | ●                 | ●      | ●    |
| MAYHTR1A10BKR* |                   | 10                | 55.6                  | 48.7       | 60                                  | 50    | ●                 | ●      | ●    |
| MAYHTR1A15BKR* |                   | 15                | 55.6/26               | 48.7/22.8  | 60/30                               | 50/25 | ●                 | ●      | ●    |
| MAYHTR1A20BKR* |                   | 20                | 55.6/52.1             | 48.7/45.1  | 60/60                               | 50/50 | ●                 | ●      | ●    |

\* Heat kit suitable for AHU 4-way position installation[● means available, --means not available].  
Ampacities for MCA and Fuse/breaker including the blower motor.  
HEATER MODEL NUMBER DIGITS '\*': A , B

## Electric Heater Kits

| NO. | Kit#           | Description                             | Ref. Air Handler use |
|-----|----------------|-----------------------------------------|----------------------|
| 1   | MAYHTR1A05BKR* | 5kW Heat Strip                          | 18,24,30,36,42,48,60 |
| 2   | MAYHTR1A08BKR* | 7.5kW Heat Strip                        | 18,24,30,36,42,48,60 |
| 3   | MAYHTR1A10BKR* | 10kW Heat Strip                         | 18,24,30,36,42,48,60 |
| 4   | MAYHTR1A15BKR* | 15kW Heat Strip, Double Breaker's panel | 36,42,48,60          |
| 5   | MAYHTR1A20BKR* | 20kW Heat Strip, Double Breaker's panel | 36,42,48,60          |

HEATER MODEL NUMBER DIGITS \*\*: A , B

## 5.0 AIRFLOW PERFORMANCE (AMERISTAR AIR HANDLERS ARE SUITABLE FOR MOBILE HOME APPLICATIONS)

Airflow performance data is based on cooling performance with a coil and no filter in place. Select performance table for appropriate unit size external static applied to unit allows operation within the minimum and maximum limits shown in table below for cooling.

### IMPORTANT:

1. When model 42 used for mobile home, you need to ensure that the air volume is not less than 1335 CFM.
2. When model 48 used for mobile home, you need to ensure that the air volume is not less than 1584 CFM.

### AIRFLOW PERFORMANCE DATA

| Model Number | Motor Speed |       | CFM(Watts)                                |          |           |          |          |          |          |          |          |          |
|--------------|-------------|-------|-------------------------------------------|----------|-----------|----------|----------|----------|----------|----------|----------|----------|
|              |             |       | External Static Pressure-Inches W.C.[kPa] |          |           |          |          |          |          |          |          |          |
|              |             |       | 0[0]                                      | 0.1[.02] | 0.16[.04] | 0.2[.05] | 0.3[.07] | 0.4[.10] | 0.5[.12] | 0.6[.15] | 0.7[.17] | 0.8[.20] |
| 18           | Low         | CFM   | 551                                       | 509      | 478       | 462      | 393      | 345      | 280      | -        | -        | -        |
|              |             | RPM   | 440                                       | 518      | 576       | 595      | 679      | 726      | 781      | -        | -        | -        |
|              |             | Watts | 122.4                                     | 120      | 118.2     | 116.8    | 116.3    | 109.9    | 106.2    | -        | -        | -        |
|              |             | Amps  | 0.62                                      | 0.62     | 0.62      | 0.62     | 0.61     | 0.61     | 0.6      | -        | -        | -        |
|              | Middle      | CFM   | 661                                       | 622      | 596       | 577      | 506      | 443      | 400      | -        | -        | -        |
|              |             | RPM   | 518                                       | 580      | 618       | 640      | 731      | 770      | 812      | -        | -        | -        |
|              |             | Watts | 145.2                                     | 143      | 141.6     | 140.8    | 136.3    | 133.6    | 131.2    | -        | -        | -        |
|              |             | Amps  | 0.69                                      | 0.68     | 0.68      | 0.68     | 0.67     | 0.67     | 0.66     | -        | -        | -        |
|              | High        | CFM   | 861                                       | 807      | 765       | 729      | 682      | 634      | 590      | 550      | 487      | 400      |
|              |             | RPM   | 693                                       | 720      | 758       | 787      | 831      | 871      | 894      | 911      | 940      | 975      |
|              |             | Watts | 265                                       | 258      | 255       | 251.3    | 243.6    | 235.4    | 232      | 229.5    | 224.4    | 217.4    |
|              |             | Amps  | 1.21                                      | 1.20     | 1.19      | 1.19     | 1.18     | 1.17     | 1.16     | 1.15     | 1.14     | 1.12     |
| 24           | Low         | CFM   | 646                                       | 623      | 602       | 592      | 553      | 506      | 453      | -        | -        | -        |
|              |             | RPM   | 528                                       | 591      | 628       | 650      | 728      | 790      | 840      | -        | -        | -        |
|              |             | Watts | 169.4                                     | 166.4    | 164.5     | 163      | 157.5    | 151.3    | 146.3    | -        | -        | -        |
|              |             | Amps  | 0.79                                      | 0.78     | 0.78      | 0.78     | 0.77     | 0.76     | 0.75     | -        | -        | -        |
|              | Middle      | CFM   | 815                                       | 802      | 786       | 771      | 733      | 681      | 613      | -        | -        | -        |
|              |             | RPM   | 658                                       | 701      | 732       | 743      | 790      | 841      | 888      | -        | -        | -        |
|              |             | Watts | 218                                       | 216.6    | 215.9     | 214.4    | 211.4    | 207.4    | 203      | -        | -        | -        |
|              |             | Amps  | 0.96                                      | 0.95     | 0.95      | 0.95     | 0.94     | 0.93     | 0.92     | -        | -        | -        |
|              | High        | CFM   | 1043                                      | 993      | 965       | 947      | 920      | 866      | 780      | 695      | 607      | 515      |
|              |             | RPM   | 741                                       | 769      | 788       | 802      | 837      | 876      | 919      | 956      | 986      | 1014     |
|              |             | Watts | 286.8                                     | 279.4    | 275.6     | 271.3    | 267.5    | 265.4    | 262.6    | 256.4    | 250.1    | 243      |
|              |             | Amps  | 1.30                                      | 1.28     | 1.26      | 1.25     | 1.24     | 1.22     | 1.2      | 1.18     | 1.17     | 1.15     |

| Model Number | Motor Speed | CFM(Watts)                                |      |          |           |          |          |          |          |          |          |          |
|--------------|-------------|-------------------------------------------|------|----------|-----------|----------|----------|----------|----------|----------|----------|----------|
|              |             | External Static Pressure-Inches W.C.[kPa] |      |          |           |          |          |          |          |          |          |          |
|              |             |                                           | 0[0] | 0.1[.02] | 0.16[.04] | 0.2[.05] | 0.3[.07] | 0.4[.10] | 0.5[.12] | 0.6[.15] | 0.7[.17] | 0.8[.20] |
| 30           | Low         | CFM                                       | 962  | 913      | 886       | 870      | 813      | 750      | 690      | -        | -        | -        |
|              |             | RPM                                       | 729  | 754      | 798       | 803      | 858      | 873      | 902      | -        | -        | -        |
|              |             | Watts                                     | 315  | 304      | 298       | 292      | 280      | 269      | 258      | -        | -        | -        |
|              |             | Amps                                      | 1.39 | 1.35     | 1.32      | 1.3      | 1.26     | 1.22     | 1.19     | -        | -        | -        |
|              | Middle      | CFM                                       | 1094 | 1043     | 1012      | 988      | 927      | 861      | 788      | -        | -        | -        |
|              |             | RPM                                       | 809  | 845      | 859       | 885      | 911      | 932      | 954      | -        | -        | -        |
|              |             | Watts                                     | 334  | 325      | 319       | 315      | 303      | 290      | 279      | -        | -        | -        |
|              |             | Amps                                      | 1.46 | 1.43     | 1.4       | 1.38     | 1.35     | 1.3      | 1.26     | -        | -        | -        |
|              | High        | CFM                                       | 1374 | 1311     | 1268      | 1240     | 1164     | 1084     | 996      | 910      | 828      | 744      |
|              |             | RPM                                       | 910  | 937      | 947       | 954      | 972      | 989      | 1004     | 1018     | 1033     | 1043     |
|              |             | Watts                                     | 440  | 427      | 419       | 413      | 398      | 381      | 366      | 352      | 340      | 331      |
|              |             | Amps                                      | 1.94 | 1.9      | 1.86      | 1.84     | 1.79     | 1.74     | 1.68     | 1.63     | 1.6      | 1.56     |
| 36           | Low         | CFM                                       | 1129 | 1088     | 1061      | 1040     | 988      | 941      | 819      | -        | -        | -        |
|              |             | RPM                                       | 642  | 675      | 706       | 732      | 771      | 817      | 853      | -        | -        | -        |
|              |             | Watts                                     | 322  | 312      | 306       | 301      | 289      | 269      | 254      | -        | -        | -        |
|              |             | Amps                                      | 1.54 | 1.52     | 1.51      | 1.5      | 1.47     | 1.42     | 1.39     | -        | -        | -        |
|              | Middle      | CFM                                       | 1317 | 1268     | 1237      | 1217     | 1157     | 1111     | 1027     | -        | -        | -        |
|              |             | RPM                                       | 776  | 810      | 841       | 874      | 905      | 935      | 966      | -        | -        | -        |
|              |             | Watts                                     | 360  | 354      | 348       | 345      | 335      | 323      | 309      | -        | -        | -        |
|              |             | Amps                                      | 1.69 | 1.67     | 1.66      | 1.65     | 1.63     | 1.6      | 1.57     | -        | -        | -        |
|              | High        | CFM                                       | 1643 | 1581     | 1544      | 1518     | 1446     | 1356     | 1261     | 1123     | 915      | 812      |
|              |             | RPM                                       | 868  | 883      | 895       | 906      | 931      | 955      | 978      | 1013     | 1028     | 1050     |
|              |             | Watts                                     | 463  | 451      | 443       | 438      | 429      | 415      | 401      | 371      | 356      | 343      |
|              |             | Amps                                      | 2.22 | 2.2      | 2.18      | 2.17     | 2.14     | 2.12     | 2.09     | 2.03     | 1.99     | 1.96     |
| 42           | Low         | CFM                                       | 1239 | 1203     | 1178      | 1161     | 1117     | 1070     | 1000     | -        | -        | -        |
|              |             | RPM                                       | 738  | 775      | 797       | 808      | 844      | 872      | 905      | -        | -        | -        |
|              |             | Watts                                     | 396  | 385      | 376       | 371      | 360      | 345      | 327      | -        | -        | -        |
|              |             | Amps                                      | 1.73 | 1.68     | 1.64      | 1.62     | 1.57     | 1.52     | 1.44     | -        | -        | -        |
|              | Middle      | CFM                                       | 1480 | 1431     | 1399      | 1379     | 1319     | 1259     | 1187     | -        | -        | -        |
|              |             | RPM                                       | 843  | 865      | 882       | 893      | 915      | 937      | 959      | -        | -        | -        |
|              |             | Watts                                     | 430  | 416      | 407       | 401      | 388      | 375      | 359      | -        | -        | -        |
|              |             | Amps                                      | 1.87 | 1.81     | 1.77      | 1.75     | 1.69     | 1.63     | 1.57     | -        | -        | -        |
|              | High        | CFM                                       | 1738 | 1682     | 1639      | 1618     | 1548     | 1477     | 1378     | 1286     | 1042     | 908      |
|              |             | RPM                                       | 921  | 941      | 949       | 955      | 970      | 985      | 1002     | 1016     | 1042     | 1059     |
|              |             | Watts                                     | 508  | 493      | 486       | 478      | 460      | 445      | 431      | 412      | 373      | 354      |
|              |             | Amps                                      | 2.22 | 2.15     | 2.12      | 2.08     | 2.01     | 1.94     | 1.89     | 1.81     | 1.65     | 1.58     |
| 48           | Low         | CFM                                       | 1471 | 1427     | 1395      | 1374     | 1316     | 1247     | 1180     | -        | -        | -        |
|              |             | RPM                                       | 694  | 732      | 753       | 769      | 803      | 833      | 864      | -        | -        | -        |
|              |             | Watts                                     | 381  | 376      | 372       | 370      | 364      | 357      | 349      | -        | -        | -        |
|              |             | Amps                                      | 1.66 | 1.64     | 1.63      | 1.62     | 1.6      | 1.58     | 1.55     | -        | -        | -        |
|              | Middle      | CFM                                       | 1729 | 1678     | 1646      | 1625     | 1558     | 1491     | 1402     | -        | -        | -        |
|              |             | RPM                                       | 790  | 817      | 833       | 845      | 876      | 898      | 920      | -        | -        | -        |
|              |             | Watts                                     | 485  | 477      | 473       | 470      | 460      | 451      | 440      | -        | -        | -        |
|              |             | Amps                                      | 1.86 | 1.84     | 1.82      | 1.81     | 1.78     | 1.75     | 1.71     | -        | -        | -        |
|              | High        | CFM                                       | 2045 | 1992     | 1951      | 1928     | 1847     | 1763     | 1677     | 1563     | 1450     | 1317     |
|              |             | RPM                                       | 895  | 920      | 932       | 938      | 956      | 972      | 987      | 1002     | 1015     | 1030     |
|              |             | Watts                                     | 641  | 627      | 617       | 612      | 596      | 582      | 566      | 546      | 528      | 507      |
|              |             | Amps                                      | 2.86 | 2.82     | 2.8       | 2.78     | 2.73     | 2.68     | 2.64     | 2.57     | 2.52     | 2.45     |
| 60           | Low         | CFM                                       | 1786 | 1740     | 1709      | 1688     | 1630     | 1562     | 1489     | -        | -        | -        |
|              |             | RPM                                       | 830  | 843      | 849       | 856      | 890      | 921      | 942      | -        | -        | -        |
|              |             | Watts                                     | 584  | 569      | 560       | 552      | 536      | 516      | 497      | -        | -        | -        |
|              |             | Amps                                      | 2.64 | 2.59     | 2.55      | 2.54     | 2.58     | 2.42     | 2.37     | -        | -        | -        |
|              | Middle      | CFM                                       | 2140 | 2071     | 2039      | 2006     | 1932     | 1799     | 1677     | -        | -        | -        |
|              |             | RPM                                       | 917  | 930      | 938       | 943      | 957      | 970      | 990      | -        | -        | -        |
|              |             | Watts                                     | 645  | 630      | 623       | 617      | 602      | 585      | 569      | -        | -        | -        |
|              |             | Amps                                      | 2.87 | 2.81     | 2.78      | 2.76     | 2.71     | 2.65     | 2.59     | -        | -        | -        |
|              | High        | CFM                                       | 2357 | 2276     | 2225      | 2188     | 2100     | 2004     | 1902     | 1764     | 1554     | 1393     |
|              |             | RPM                                       | 964  | 976      | 982       | 990      | 1000     | 1012     | 1022     | 1032     | 1042     | 1063     |
|              |             | Watts                                     | 754  | 733      | 718       | 710      | 693      | 673      | 650      | 630      | 607      | 575      |
|              |             | Amps                                      | 3.34 | 3.27     | 3.22      | 3.19     | 3.12     | 3.05     | 2.98     | 2.91     | 2.83     | 2.71     |

The air distribution system has the greatest effect on airflow. The duct system is totally controlled by the contractor. For this reason, the contractor should use only industry-recognized procedures.

Heat pump systems require a specified airflow. Each ton of cooling requires between 350 and 450 cubic feet of air per minute (CFM), or 400 CFM nominally.

Duct design and construction should be carefully done. System performance can be lowered dramatically through bad planning or workmanship.

Air supply diffusers must be selected and located carefully. They must be sized and positioned to deliver treated air along the perimeter of the space. If they are too small for their intended airflow, they become noisy. If they are not located properly, they cause drafts. Return air grilles must be properly sized to carry air back to the blower. If they are too small, they also cause noise.

The installers should balance the air distribution system to ensure proper quiet airflow to all rooms in the home. This ensures a comfortable living space.

An air velocity meter or airflow hood can give a reading of system CFM.

## 6.0 DUCTWORK

Field ductwork must comply with the National Fire Protection Association NFPA 90A, NFPA 90B and any applicable local ordinance.



### WARNING

Do not, under any circumstances, connect return ductwork to any other heat producing device such as fireplace insert, stove, etc. Unauthorized use of such devices may result in fire, carbon monoxide poisoning, explosion, personal injury or property damage.

Sheet metal ductwork run in unconditioned spaces must be insulated and covered with a vapor barrier. Fibrous ductwork may be used if constructed and installed in accordance with SMACNA Construction Standard on Fibrous Glass Ducts. Ductwork must comply with National Fire Protection Association as tested by U/L Standard 181 for Class I Air Ducts. Check local codes for requirements on ductwork and insulation.

- Duct system must be designed within the range of external static pressure the unit is designed to operate against. It is important that the system airflow be adequate. Make sure supply and return ductwork, grills, special filters, accessories, etc. are accounted for in total resistance. See airflow performance tables in this manual.
- Design the duct system in accordance with "ACCA" Manual "D" Design for Residential Winter and Summer Air Conditioning and Equipment Selection. Latest editions are available from: "ACCA" Air Conditioning Contractors of America, 1513 16th Street, N.W., Washington, D.C. 20036. If duct system incorporates flexible air duct, be sure pressure drop information (straight length plus all turns) shown in "ACCA" Manual "D" is accounted for in system.
- Supply plenum is attached to the 3/4" duct flanges supplied with the unit. Attach flanges around the blower outlet.

**IMPORTANT:** If an elbow is included in the plenum close to the unit, it must not be smaller than the dimensions of the supply duct flange on the unit.

- **IMPORTANT:** The front flange on the return duct if connected to the blower casing must not be screwed into the area where the power wiring is located. Drills or sharp screw points can damage insulation on wires located inside unit.
- Secure the supply and return ductwork to the unit flanges, using proper fasteners for the type of duct used and tape the duct-to-unit joint as required to prevent air leaks.

## 7.0 REFRIGERANT CONNECTIONS

Keep the coil connections sealed until refrigerant connections are made. See the Installation Instructions for the outdoor unit for details on line sizing, tubing installation, and charging information.

The coil comes from the factory with a nitrogen holding charge. Evacuate the system before charging with refrigerant.

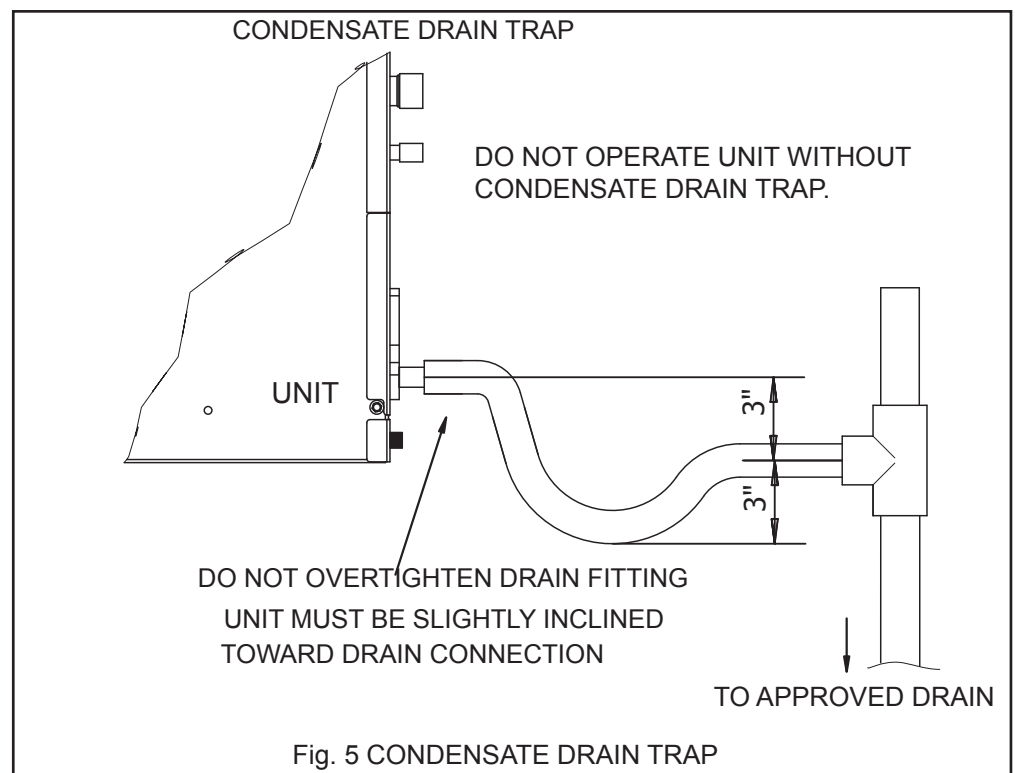
Install refrigerant tubing so that it does not block service access to the front of the unit.

Nitrogen should flow through the refrigerant lines while brazing.

Use a brazing shield to protect the cabinet's paint and a wet rag to protect the rubber grommet from being damaged by torch flames. After the refrigerant connections are made, seal the gap around the connections with pressure sensitive gasket.

## 7.1 CONDENSATE DRAIN TUBING

Consult local codes for specific requirements.



### IMPORTANT:

1. When making drain fitting connections to the drain pan, use a thin layer of Teflon paste, silicone or Teflon tape and install, hand tighten.
2. When making drain fitting connections to drain pan, do not overtighten. Overtightening fittings can split pipe connections on the drain pan.

- Install drain lines so they do not block service access to front of the unit. Minimum clearance of 24 inches is required for filter, coil or blower removal and service access.
- Make sure unit is level or pitched slightly toward primary drain connection so that water will drain completely from the pan. (See Fig. 5)
- Do not reduce drain line size less than connection size provided on condensate drain pan.
- All drain lines must be pitched downward away from the unit a minimum of 1/4" per foot of line to ensure proper drainage.
- Do not connect condensate drain line to a closed or open sewer pipe. Run condensate to an open drain or run line to a safe outdoor area.
- The drain line should be insulated where necessary to prevent sweating and damage due to condensate forming on the outside surface of the line.
- Make provisions for disconnecting and cleaning of the primary drain line should it become necessary. Install a 3 inch trap in the primary drain line as close to the unit as possible. Make sure that the top of the trap is below connection to the drain pan to allow complete drainage of pan (See Fig. 5).
- Auxiliary drain line should be run to a place where it will be noticeable if it becomes operational. Homeowner should be warned that a problem exists if water should begin running from the auxiliary drain line.
- Plug the unused drain connection with the plugs provided in the parts bag, using a thin layer of teflon paste, silicone or teflon tape to form a water tight seal.
- Test condensate drain pan and drain line after installation is complete. Pour water into drain pan, enough to fill drain trap and line. Check to make sure drain pan is draining completely, no leaks are found in drain line fittings, and water is draining from the termination of the primary drain line.

## 8.0 AIR FILTER (not factory-installed)

- External filter or other means of filtration is required. Units should be sized for a maximum of 300 feet/min. air velocity or what is recommended for the type filter installed.

Filter application and placement are critical to airflow, which may affect the heating and cooling system performance. Reduced airflow can shorten the life of the system's major components, such as motor, limits, elements, heat relays, evaporator coil or compressor.

If adding high efficiency filters or electronic air filtration systems, it is very important that the air flow is not reduced. If air flow is reduced the overall performance and efficiency of the unit will be reduced. It is strongly recommended that a professional installation technician is contacted to ensure installation of these such filtration systems are installed correctly.

**IMPORTANT:** DO NOT DOUBLE FILTER THE RETURN AIR DUCT SYSTEM. DO NOT FILTER THE SUPPLY AIR DUCT SYSTEM. THIS WILL CHANGE THE PERFORMANCE OF THE UNIT AND REDUCE AIRFLOW .

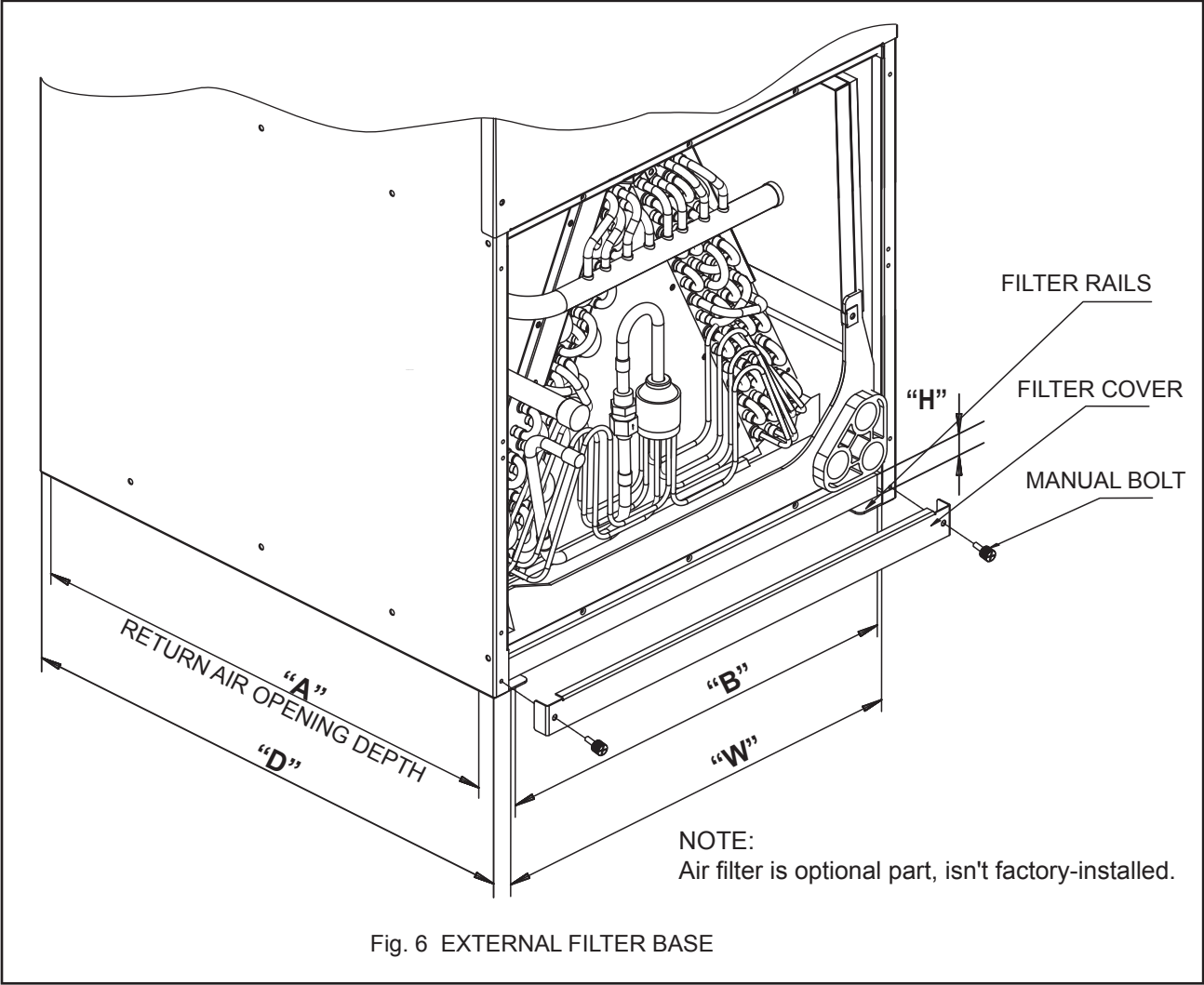


### WARNING

Do not operate the system without filters. A portion of the dust entrained in the air may temporarily lodge in the duct runs and at the supply registers. Any circulated dust particles could be heated and charred by contact with the air handler elements. This residue could soil ceilings, walls, drapes, carpets and other articles in the house.

Soot damage may occur with filters in place, when certain types of candles, oil lamps or standing pilots are burned.

# 9.0 FILTER INSTALLATION DIMENSIONS

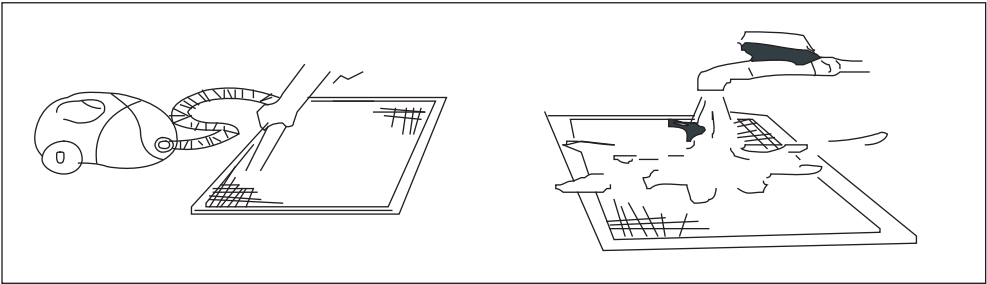


## DIMENSIONAL DATA

| MODEL    | FILTER SIZE<br>IN [mm] | "W" IN [mm] | "D" IN [mm] | "H" IN [mm] | Return width<br>"A" IN | Return length<br>"B" IN |
|----------|------------------------|-------------|-------------|-------------|------------------------|-------------------------|
| 18/24/30 | 16X20[406X508]         | 16.8[426]   | 20.4[518]   | 1[25.4]     | 19.6                   | 14.8                    |
| 36/42    | 18X20[457X508]         | 18.3[466]   | 21.6[548]   | 1[25.4]     | 20.8                   | 16.3                    |
| 48/60    | 20X22[508X559]         | 20.7[526]   | 23.9[608]   | 1[25.4]     | 23                     | 18.8                    |

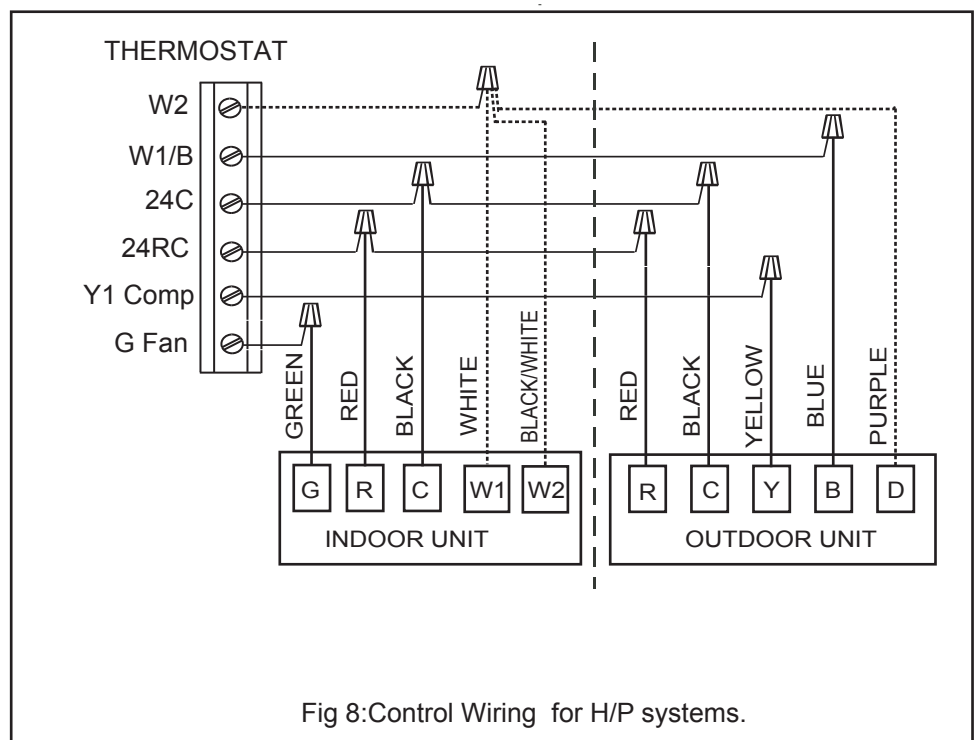
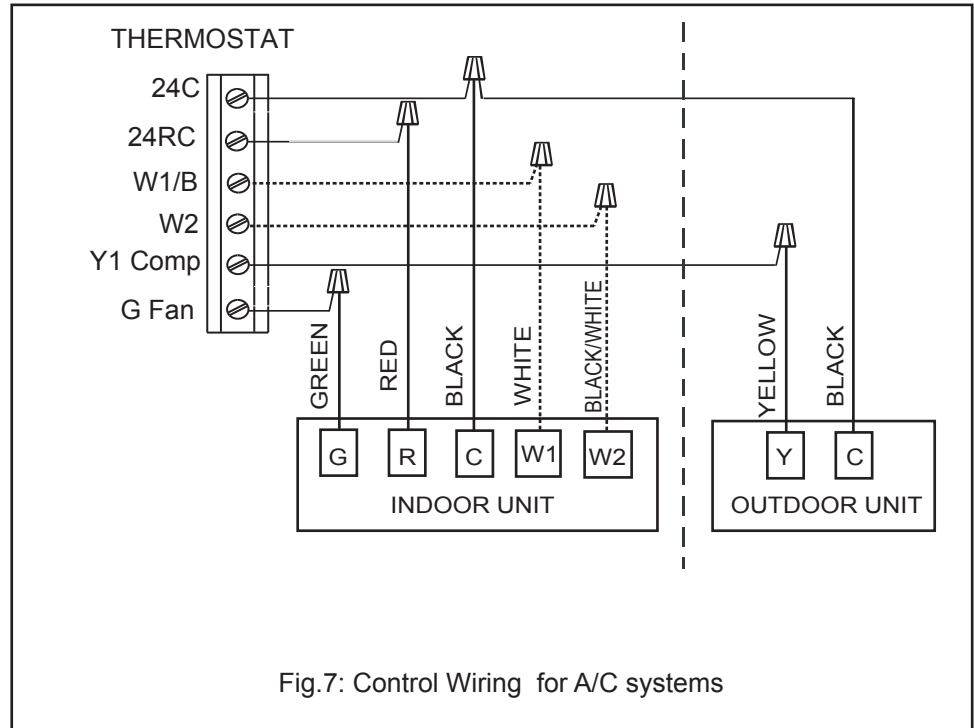
### • AIR FILTER REMOVAL

- 1.Remove bolts manually, remove air filter recover, see in Fig 6;
2. Hold the edge of the air filter and extract out .
3. Clean the air filter (Vacuum cleaner or pure water may be used to clean the air filter. If the dust accumulation is too heavy, use soft brush and mild detergent to clean it and dry out in cool place) .



## 10.0 WIRING DIAGRAM

1. To avoid the electrical shock, please connect the air conditioner with the ground lug. The main power plug in the air conditioner has been joined with the ground wiring, please don't change it freely.
2. The power socket is used as the air conditioner specially.
3. Don't pull the power wiring hard.
4. When connecting the air conditioner with the ground, observe the local codes.
5. If necessary, use the power fuse or the circuit, breaker or the corresponding scale ampere.



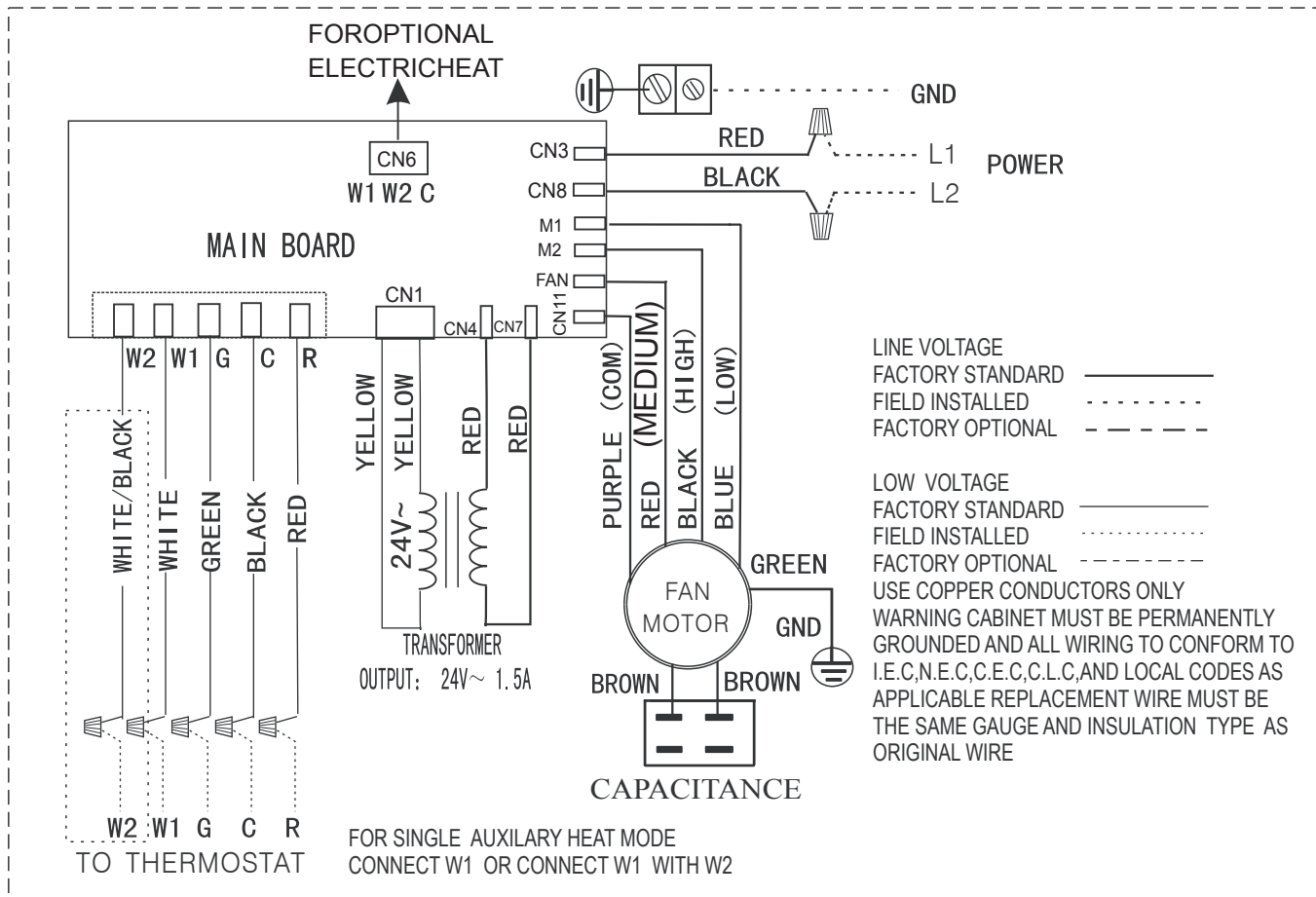


Fig.9: Indoor Unit Wiring Diagram for A/C systems and H/P systems.

Note:Description of fan speed switch

- 1.Default as medium speed of factory settings.
- 2.High speed wiring: Switch to high speed (black wire) and connect with FAN terminal, while medium speed (red wire) connect with M2 terminal.
- 3.Low speed wiring: Switch to low speed (blue wire) and connect with FAN terminal, while medium speed (red wire) connect with M1 terminal.

| Terminal<br>Fan speed | Fan   | M1   | M2    |
|-----------------------|-------|------|-------|
| Medium                | Red   | Blue | Black |
| High                  | Black | Blue | Red   |
| Low                   | Blue  | Red  | Black |

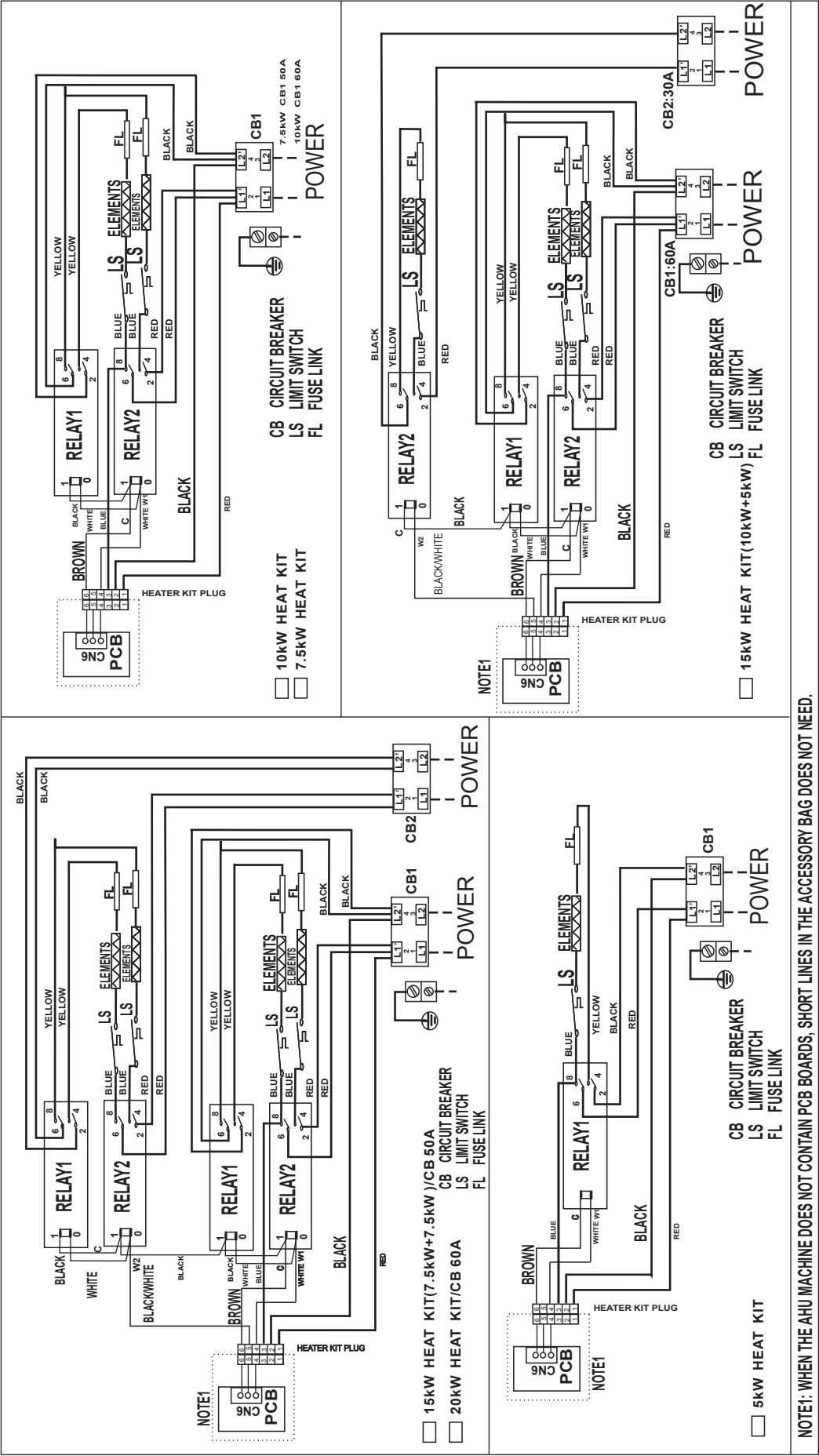


Fig. 10: Indoor Unit Wiring Diagram for Electric Heat.

## ELECTRIC WIRING GAUGE

Wiring gauge for A/C systems

| Model(Btu/h)   |                             |                    | 18/24          | 30/36<br>/42 | 48  | 60  |
|----------------|-----------------------------|--------------------|----------------|--------------|-----|-----|
| Power          |                             | Phase              | Single         |              |     |     |
|                |                             | Voltage/frequency  | 208/230V, 60Hz |              |     |     |
| Lines<br>Gauge | Input Current Fuse          | Indoor unit (A)    | 15A            | 15A          | 15A | 15A |
|                | Indoor Unit Power Line      | Line Quantity      | 3              | 3            | 3   | 3   |
|                |                             | Line Diameter(AWG) | 14             | 14           | 14  | 14  |
|                | Outdoor Unit Power Line     | Line Quantity      | 3              | 3            | 3   | 3   |
|                |                             | Line Diameter(AWG) | 14             | 12           | 10  | 10  |
|                | Outdoor -Indoor Signal Line | Line Quantity      | 2              | 2            | 2   | 2   |
|                |                             | Line Diameter(AWG) | 18             | 18           | 18  | 18  |
|                | Thermostat Signal Line      | Line Quantity      | 4              | 4            | 4   | 4   |
|                |                             | Line Diameter(AWG) | 18             | 18           | 18  | 18  |

NOTE: If indoor unit has auxiliary heating already installed and a different auxiliary heating unit is required the indoor unit (A) and indoor line diameters will be different.

Wiring gauge for H/P systems

| Model(Btu/h)   |                             |                    | 18/24          | 30/36<br>/42 | 48  | 60  |
|----------------|-----------------------------|--------------------|----------------|--------------|-----|-----|
| Power          |                             | Phase              | Single         |              |     |     |
|                |                             | Voltage/frequency  | 208/230V, 60Hz |              |     |     |
| Lines<br>Gauge | Input Current Fuse          | Indoor unit (A)    | 15A            | 15A          | 15A | 15A |
|                | Indoor Unit Power Line      | Line Quantity      | 3              | 3            | 3   | 3   |
|                |                             | Line Diameter(AWG) | 14             | 14           | 14  | 14  |
|                | Outdoor Unit Power Line     | Line Quantity      | 3              | 3            | 3   | 3   |
|                |                             | Line Diameter(AWG) | 14             | 12           | 10  | 10  |
|                | Outdoor -Indoor Signal Line | Line Quantity      | 4              | 4            | 4   | 4   |
|                |                             | Line Diameter(AWG) | 18             | 18           | 18  | 18  |
|                | Thermostat Signal Line      | Line Quantity      | 5              | 5            | 5   | 5   |
|                |                             | Line Diameter(AWG) | 18             | 18           | 18  | 18  |

NOTE: If indoor unit has auxiliary heating already installed and a different auxiliary heating unit is required the indoor unit (A) and indoor line diameters will be different.

These units must be wired and installed in accordance with all National and Local Safety Codes.

# Orifice/TXV Replacement Information

**NOTE:** The orifice / TXV replacement options noted in this sheet supersede those in the installation guide. Please reference this sheet for all refrigerant metering options.

Step 1: Remove the screws and front coil panel.

Step 2: Remove the rubber plugs from the liquid and vapor lines.

Step 3: Using a back up wrench and loosen the flow assembly. Remove the factory installed piston Reassemble and tighten the flow assembly. Tighten the flow assembly to 11 ( $\pm 2$ ) ft-lb

Step 4: If replacing the preinstalled orifice with the optional orifice, ensure the replacement orifice is placed in the same orientation as the preinstalled orifice before tightening the flow assembly.

## TXV-Specific Steps

Step 5: Drill a small hole in the suction line for the TXV equalizer line. The hole must be on top as noted in the picture.

Step 6: Dry fit the TXV to the liquid line entering the coil. The valve must be in the upright position as pictured. Keep the valve as close to the coil as possible, although a short piece of field fabricated tubing may be needed. The TXV must be mounted in the CORRECT direction of flow. Place the TXV equalizer line 3 to 4 mm inside the small hole drilled in the vapor line.

Step 7: Wrap the TXV and coil panel with a wet rag to prevent overheating while brazing. Use a nitrogen flow and braze all connections.

Step 8: Allow tubing to cool and pressurize line sets with 150 PSI of nitrogen to check braze connections and flow assembly for leaks. Make repairs as needed.

Step 9: Locate and clean a straight section of the vapor line as close to the coil as possible. Use the supplied copper straps to secure the TXV sensing bulb on top of the vapor line as pictured.

Step 10: Insulate the entire vapor line and sensing bulb. It is also recommended to insulate the TXV and liquid line between the valve and coil to prevent condensation in hot humid environments.

Step 11: Replace the front coil panel and secure in place.

Step 12: Follow the steps in the installation guide for vacuum requirements and system start up procedures.

Step 13: Allow system to run for a minimum of 10 minutes in the cooling mode.

A: For fixed orifice installation: Use the superheat charging chart to obtain proper superheat based on indoor / outdoor conditions.

B: For TXV installation: Charge the system to 10 degrees of subcooling. Adjust the expansion valve to achieve 9 ( $\pm 3$  degrees) of superheat.

Step 14: Allow system to run for an additional 10 minutes to verify the subcooling and superheat readings.

DRILL A SMALL HOLE IN VAPOR LINE FOR EQUALIZER TUBE

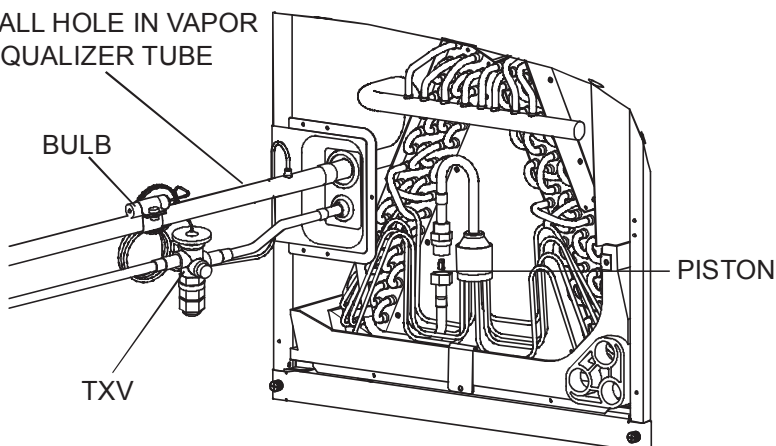


TABLE 1: The optional TXV Kit means the recommended model for field installation.

| Condensing Unit |                 | Air Handle / A-Coil | Piston Size | Optional TXV Kit | SEER | If Preinstalled Piston Replace Required |
|-----------------|-----------------|---------------------|-------------|------------------|------|-----------------------------------------|
| AC              | M4AC3018B1000NA | M4AH3018B1000AA     | C050        | MAYTXVACAC1818AA | 13   | YES                                     |
|                 | M4AC3024B1000NA | M4AH3024B1000AA     | C056        | MAYTXVACHP1830AA | 13   | NO                                      |
|                 | M4AC3030B1000NA | M4AH3030B1000AA     | C063        | MAYTXVACHP1830AA | 13   | YES                                     |
|                 | M4AC3036B1000NA | M4AH3036B1000AA     | C071        | MAYTXVACHP3642AA | 13   | YES                                     |
|                 | M4AC3042B1000NA | M4AH3042B1000AA     | C072        | MAYTXVACHP3642AA | 13   | YES                                     |
|                 | M4AC3048B1000NA | M4AH3048B1000AA     | C078        | MAYTXVACHP4848AA | 13   | YES                                     |
|                 | M4AC3060B1000NA | M4AH3060B1000AA     | C090        | MAYTXVACHP6060AA | 13   | YES                                     |
|                 | M4AC4060B1000NA | M4AH3060B1000AA     | C102        | MAYTXVACHP6060AA | 14   | YES                                     |
|                 |                 |                     |             |                  |      |                                         |
| HP              | M4HP3018A1000AA | M4AH3018B1000AA     | C051        | MAYTXVACHP1830AA | 13   | NO                                      |
|                 | M4HP3024A1000AA | M4AH3024B1000AA     | C056        | MAYTXVACHP1830AA | 13   | NO                                      |
|                 | M4HP3030A1000AA | M4AH3030B1000AA     | C061        | MAYTXVACHP1830AA | 13   | NO                                      |
|                 | M4HP3036A1000AA | M4AH3036B1000AA     | C066        | MAYTXVACHP3642AA | 13   | NO                                      |
|                 | M4HP3042A1000AA | M4AH3042B1000AA     | C075        | MAYTXVACHP3642AA | 13   | NO                                      |
|                 | M4HP3048A1000AA | M4AH3048B1000AA     | C083        | MAYTXVACHP4848AA | 13   | NO                                      |
|                 | M4HP3060A1000AA | M4AH3060B1000AA     | C096        | MAYTXVACHP6060AA | 13   | NO                                      |
|                 | M4HP4060A1000A* | M4AH3060B1000AA     | C096        | MAYTXVACHP6060AA | 14 • | YES                                     |

Note: \* represents A and B.

- To achieve 14 SEER, remove piston and install a TXV valve (model MAYTXVACHP6060AA).

## Base Limited Warranty

### Single Phase R-410A Outdoor Units, Single Phase R-22 Outdoor Units, Air Handlers, Furnaces, Packaged Units, Cased Coils, Ductless and Specific Terms for Commercial Applications.

Subject to the terms and conditions of this limited warranty, Ingersoll Rand ("Company") extends a limited warranty against manufacturing defects for the product(s) identified in **Table 1** attached hereto ("Products") that are installed in a residential/multi-family application (personal, family or household purposes) under normal use and maintenance in the United States and Canada.

In order to maximize the available benefits under this limited warranty, the Purchaser (as defined below) should read it in its entirety. All repairs of Product parts covered under this limited warranty must be made with authorized service parts and by a licensed HVAC service provider. Additionally, commercial applications are treated differently under this limited warranty as stated in **Table 1** attached hereto. For purposes of this limited warranty, "commercial applications" shall mean any application other than for personal, family, or household use.

**TERM:** The limited warranty period for Products is as stated in **Table 1** attached hereto. If the Purchaser properly registers the Products, the limited warranty period shall be extended as stated in **Table 1** attached hereto. Regardless of registration, the Commencement Date for a limited warranty period shall be the date that the original installation is complete and all Product start-up procedures have been properly completed and verified by an installer's invoice. If the installation and start-up date cannot be verified by the installer's invoice, the Commencement Date shall be sixty (60) days after the factory manufacture date which is verified by the Product serial number. Where a Product is installed in a newly constructed home, the Commencement Date is the date the Purchaser purchased the residence from the builder. Proof of Product purchase, installation, and/or closing date of the residence may be required to confirm the Commencement Date.

The installation of Product replacement parts under this limited warranty shall not extend the original warranty period. The warranty period for any Product part replaced under this limited warranty is the applicable warranty period remaining under the original Product warranty.

**WHO IS COVERED:** This limited warranty is provided only to the original owner and his or her spouse ("Purchaser") of the residence where the Products are originally installed. This warranty is not transferable except according to terms stated on the applicable website identified below under Registration Requirements. Company has the right to request any and all proof of Product purchase or installation and/or closing date of the residence.

**WHAT COMPANY WILL DO:** Company may request proof of Product purchase and/or installation in order to provide Product parts under this limited warranty. As Company's only responsibility and Purchaser's only remedy under this limited warranty, Company will furnish a replacement part to the licensed HVAC service provider, without charge for the part only, to replace any Product part that fails due to a manufacturing defect under normal use and maintenance. The Purchaser must pay for any and all shipping and handling charges and other costs of warranty service for the replacement part. If a Product part is not available, Company will, at its option, provide a free suitable substitute part or provide a credit in the amount of the then factory selling price for a new suitable substitute part to be used by the Purchaser towards the retail purchase price of a new Company product. Any new Product purchase shall be at Purchaser's sole cost and expense including, but not limited to, all shipping, removal, and installation costs and expenses.

**REGISTRATION REQUIREMENTS:** All Products must be properly registered online by the Purchaser within sixty (60) days after the Commencement Date to receive the registered limited warranty terms. To register online, go to:

<http://www.ameristarac.com>

and click "Begin Online Registration." If a Purchaser does not register within this stated time period, the base limited warranty terms shall apply.

**ELIGIBILITY REQUIREMENTS:** The following items are required in order for the Products to be covered under this limited warranty:

- The Products must be in the same location where they were originally installed.
- The Products must be properly installed, operated, and maintained by a licensed HVAC service provider in accordance with the Product specifications or installation, operation, and maintenance instructions provided by Company with each Product. Failure to conform to such specifications and/or instructions shall void this limited warranty. Company may request written documentation showing the proper preventative maintenance.
- All Product parts replaced by Company under this limited warranty must be given to the servicing provider for return to Company.
- Air handlers, air conditioners, heat pumps, cased or uncased coils, stand-alone furnaces, and packaged units must be part of an Air Conditioning, Heating, and Refrigeration Institute rated and matched system or a specification in a Company provided bulletin or otherwise approved in writing by a Company authorized representative.

**EXCLUSIONS:** The following are not covered by this limited warranty:

- Labor costs including, but not limited to, costs for diagnostic calls or the removal and reinstallation of Products and/or Product parts.
- Shipping and freight expenses required to ship Product replacement parts.
- Failures, defects, or damage (including, but not limited to, any loss of data or property) caused by (1) any third party product, service, or system connected or used in conjunction with the Products; (2) any use that is not designed or intended for the Products; (3) modification, alteration, abuse, misuse, negligence, or accident; (4) improper storage, installation, maintenance, or operation including, but not limited to, operation of electrical equipment at voltages other than the range specified on the Product nameplate; (5) any use in violation of written instructions or specifications provided by Company; (6) any acts of God including, but not limited to, fire, water, storms, lightning, or earthquakes; or any theft or riots; or (7) a corrosive atmosphere or contact with corrosive materials such as, but not limited to, chlorine, fluorine, salt, sulfur, recycled waste water, urine, fertilizers, rust, or other damaging substances or chemicals.
- Products purchased direct including, but not limited to, Internet or auction purchases and purchases made on an uninstalled basis.
- Cabinets or cabinet pieces that do not affect product performance, air filters, refrigerant, refrigerant line sets, belts, wiring, fuses, surge protection devices, non-factory installed driers, and Product accessories.
- Increased utility usage costs.

**REFRIGERANT POLICY:** (1) **Manufacturer-Installed Refrigerant:** Beginning on January 1, 2010, R-22 refrigerant will no longer be used as a manufacturer-installed refrigerant as required by federal regulation. All Products with manufacturer-installed refrigerant will include R410-A refrigerant. Any and all expenses or costs associated with replacing Product parts that are not R-410A compatible will not be covered by the terms and conditions of this limited warranty. (2) **Non-Manufacturer installed Refrigerant:** For Products manufactured and sold by the Company without refrigerant, only manufacturer approved and genuine alternate refrigerants shall be used. The use of contaminated, counterfeit, non-genuine, or non-manufacturer approved alternate refrigerant will void this limited warranty. (3) All Products: Products include a liquid line filter drier which must be replaced when a compressor replacement is necessary. A suction line filter drier must be added for compressors defined as burnouts and failure to do so will void this warranty. Non-approved refrigerant and/or non-approved refrigerant system additives including, but not limited to dyes will void this limited warranty.

#### ADDITIONAL TERMS:

THIS LIMITED WARRANTY AND LIABILITY SET FORTH HEREIN ARE IN LIEU OF ALL OTHER WARRANTIES AND LIABILITIES, WHETHER IN CONTRACT OR IN NEGLIGENCE, EXPRESS OR IMPLIED, IN LAW OR IN FACT. THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE LIMITED TO THE DURATION OF THE APPLICABLE PRODUCT WARRANTY. COMPANY DOES NOT AUTHORIZE ANY PERSON TO CREATE FOR IT ANY OBLIGATION OR LIABILITY IN CONNECTION WITH THE PRODUCTS.

NOTWITHSTANDING ANYTHING IN THIS LIMITED WARRANTY TO THE CONTRARY, COMPANY SHALL NOT BE LIABLE FOR ANY INCIDENTAL, CONSEQUENTIAL, INDIRECT, SPECIAL AND/OR PUNITIVE DAMAGES, WHETHER BASED ON CONTRACT, WARRANTY, TORT (INCLUDING, BUT NOT LIMITED TO, STRICT LIABILITY OR NEGLIGENCE), PATENT INFRINGEMENT, OR OTHERWISE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. COMPANY'S MAXIMUM LIABILITY HEREUNDER IS LIMITED TO THE ORIGINAL PURCHASE PRICE OF THE PRODUCTS.

No action arising out of any claimed breach of this limited warranty may be brought by a Purchaser more than one (1) year after the cause of action has arisen.

This limited warranty gives you specific legal rights, and you may also have other rights as otherwise permitted by law. If this Product is considered a consumer product, please be advised that some local laws do not allow limitations on incidental or consequential damages, how long a warranty lasts based on registration, or how long an implied warranty lasts, so that the above limitations may not fully apply. Refer to your local laws for your specific rights under this limited warranty. If you have any questions regarding this limited warranty, please contact your original installation dealer, or any participating dealer, should your original installation dealer no longer be available.

## **TABLE 1: Warranty Time Periods**

**COVERAGE TERMS FOR RESIDENTIAL APPLICATIONS:** Pursuant to the Ingersoll Rand ("Company") limited warranty terms and conditions, the following Products are covered for the base time periods as stated below ("Base Limited Warranty Period"). If registered, the Base Limited Warranty Periods for certain Products will be extended as stated below ("Registered Limited Warranty Period").

### **FURNACES:**

M801P, M951P, M952V

Base Limited Warranty Period: Parts – five (5) years, Heat Exchanger – twenty (20) years.

Registered Limited Warranty Period: Parts – ten (10) years, Heat Exchanger: twenty (20) years.

### **AIR HANDLERS:**

M4AH3, M4AH4, M4AH6

Base Limited Warranty Period: Indoor Coil and Parts – five (5) years.

Registered Limited Warranty Period: Indoor Coil and Parts – ten (10) years.

TMM4, TMM5

Base Limited Warranty Period: Indoor Coil and Parts – three (3) years.

Registered Limited Warranty Period: Indoor Coil and Parts – ten (10) years.

### **ELECTRIC HEATERS, installed in M4A3/4 AIR HANDLERS:**

MAYHTR1A

Limited Warranty Period: Parts – ten (10) years.

### **SINGLE PHASE R410 OUTDOOR UNITS:**

M4AC3 (**M4AC3018, 24, 30, 36, 42, 43, 48, 60 only**), M4AC6, M4HP3, M4AC4, M4HP4, M4HP6

Base Limited Warranty Period: Compressor, Outdoor Coil, Parts: five (5) years.

Registered Limited Warranty Period: Compressor, Outdoor Coil, Parts – ten (10) years.

### **SINGLE PHASE R410 OUTDOOR UNITS:**

M4AC3 (**M4AC3017, 23, 29 only**)

Base Limited Warranty Period: Compressor, Outdoor Coil, Parts: one (1) year.

Registered Limited Warranty Period: Compressor, Outdoor Coil, Parts – five (5) years.

### **SINGLE PHASE R22 OUTDOOR UNITS:**

M2AC3, M2HP3

Base Limited Warranty Period: Compressor, Outdoor Coil, Parts - one (1) year.

Registered Limited Warranty Period: Compressor, Outdoor Coil, Parts – five (5) years.

### **CASED COILS:**

M4CXC

Base Limited Warranty Period: Coil, Parts –five (5) years.

Registered Limited Warranty Period: Coil and Parts – ten (10) years.

4MXC, 4GXC

Base Limited Warranty Period: Coil, Parts –three (3) years.

Registered Limited Warranty Period: Coil and Parts – ten (10) years.

### **PACKAGED UNITS:**

M4PH3, M4PG3, M4PH4, M4PW4, M4PG4

Base Limited Warranty Period: Compressor, Coil, Parts – five (5) years, Heat Exchanger – ten (10) years.

Registered Limited Warranty Period: Compressor, Coil, Parts – ten (10) years, Heat Exchanger – ten (10) years.

### **ELECTRIC HEATERS, installed in M4PH4 PACKAGED UNITS:**

MAYHTR1P

Limited Warranty Period: Parts – ten (10) years.

### **DUCTLESS SYSTEMS\*\*:**

**M4THS/M4MHW, M4TCS/M4MCW, M4MLW/M4TLS, M4THM:**

*Base Limited Warranty Period:*

All Parts – five (5) year.

*Registered Limited Warranty Period:*

All Parts – ten (10) years.

*\*\*There is no distinction between residential and commercial use for this Limited Warranty term and coverage.*

### **SPECIFIC TERMS FOR COMMERCIAL APPLICATIONS**

Base Limited Warranty Period Applies



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说明书材料为双胶纸100g，尺寸为大A4, 印刷颜色为黑色。

### V1.7版

产品保修卡替换

### V1.8版（E版改为F版）

产品保修卡更新、封面认证更改

### V1.9版（F版改为G版）

产品保修卡更新

### V2.0版（G版改为H版）

产品保修卡更新

### V2.1版（H版改为I版）

P12/P13页型号更改

### V2.2版（I版改J版）

保修卡更新

### V2.3版（J版改K版）

P13页第五点描述更改

### V2.4版（K版改L版）

P24页型号更改

### V2.5版（L版改M版）

P12页48型号MAYHTR1A05BKR\* 对应208V 下的最大电流25改成30

### V2.6版（M版改N版）

更新保修卡

### V2.7版（N版升O版）

P13页删除NOT  
增加

#### IMPORTANT:

1. When model 44 used for mobile home, you need to ensure that the air volume is not less than 1335 CFM.
2. When model 48 used for mobile home, you need to ensure that the air volume is not less than 1584 CFM.



